Sound Profiles of Billboard Hot 100 Hits

Group MPL09

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Datasets: Billboard Hot 100 (1958–2021) and Spotify Audio Features

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1. Introduction

This report continues the work carried out in the first phase of the project, in which two public datasets on popular music in the United States of America were explored: the Billboard Hot 100 (the weekly hits chart, with around 327 thousand rows corresponding to appearances of songs in a given week, between 1958 and 2021) and the Spotify audio features dataset (approximately 29 thousand songs, each described by attributes such as danceability, energy, valence, acousticness or tempo). In the exploratory phase, the quality of the data was assessed, and missing values in the Spotify audio features, duplicate records and impossible or poorly estimated values were identified (tempo at 0 BPM and "double counting" above ~190 BPM, incorrect durations and one case of loudness at +2.3 dB); the research questions guiding the present analysis were then formulated.

General objective. The general objective is to identify, without resorting to predefined labels, which sound profiles characterise Billboard hits and to understand how the prevalence of these profiles has evolved over the decades. To this end, the following research questions were defined:

- **Q1.** Are there natural and interpretable groups of hit songs, defined exclusively by their audio features?
- **Q2.** Does the composition of these groups vary systematically over time, reflecting the evolution of the dominant sound in the charts?

These questions follow directly from the preliminary report (Phase 1), in which the group had proposed to investigate "distinct sound patterns for each decade" and to determine "how many types of hit recipe" exist, having even anticipated weighting the observations by chart position. The seasonal dimension, also proposed in Phase 1, is addressed in a complementary manner; this report focuses on the temporal axis by decade and on the characterisation of hit profiles.

The nature of the problem, the search for latent structure in data lacking a target variable, justifies the use of unsupervised learning, specifically K-means clustering. The relationship with the temporal dimension is analysed a posteriori, by cross-tabulating the groups obtained with the decade of each observation.

2. Final data preparation

Join and unit of analysis. The two files were joined on the song_id variable through an inner join, keeping only the songs present in the Hot 100 and having audio features. The unit of analysis adopted is the "song-week" observation: each row corresponds to the presence of a song in a specific week of the chart. This is a deliberate choice, since the aim is not to describe the music catalogue in general, but rather the sound that was actually disseminated and successful in each period. A song that remained twenty weeks in the chart therefore carries more weight than another that appeared in it for a single week.

Removal of duplicates and identifiers. The audio dataset was deduplicated with dropDuplicates(song_id), ensuring a single record per song, and the redundant

columns song and performer, already present in the Billboard data, were removed. distinct() was applied to the Billboard dataset, eliminating repeated records. Identifiers and URLs were excluded from the analysis, as they have no sonic meaning.

Variable selection. From the available columns, ten continuous audio features were selected: danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo and duration (ms). The following were excluded: the chart performance variables (position, weeks on chart and peak position), as they reflect commercial success rather than sound; the redundant or poorly interpretable variables (key, mode and time_signature); and Spotify popularity, as it is a current rather than contemporaneous measure of success. The guiding criterion was to retain only variables with sonic relevance, low redundancy and clear interpretability.

Missing values and derived variable. Observations lacking any of the ten features were removed (na.drop). The decade (1950s…2020s) was derived from week_id and used exclusively as an interpretive variable; it is not part of the clustering, so that the groups are defined solely by sound.

Weighting by chart position. So that each observation reflects its importance, a weight $w = \lceil \sqrt{(101 - \text{position})} \rceil$ was assigned, ranging from 10 (top positions) to 1 (position 100). The square-root compression prevents number 1 from dominating excessively, while maintaining a top-to-bottom ratio of 10:1. Since K-means does not natively support weights, each row was replicated w times. The dataset grew from 284 136 song-week observations to 2 053 341 weighted observations.

Standardisation. The ten features have very different scales (tempo in hundreds of BPM, valence between 0 and 1, duration in hundreds of thousands of ms). Since K-means relies on Euclidean distance, the StandardScaler (mean 0 and standard deviation 1) was applied through a Pipeline. Without this step, tempo and duration would dominate the distance, cancelling out the contribution of the remaining variables.

3. Methodology

Chosen method. K-means (PySpark MLlib) was applied to the ten standardised features. This method is appropriate because the variables are continuous, the notion of sonic proximity is naturally Euclidean and the algorithm scales well to millions of weighted observations, a decisive advantage in a Big Data context, in which methods such as hierarchical clustering (of quadratic complexity) would be infeasible. The random seed was fixed (seed=42) to ensure reproducibility.

Choice of the number of clusters. Values of k from 2 to 8 were evaluated, combining two criteria: the elbow method (WSSSE) and the silhouette coefficient (computed on a 10% sample, for reasons of scale). Figure 1 presents the results.

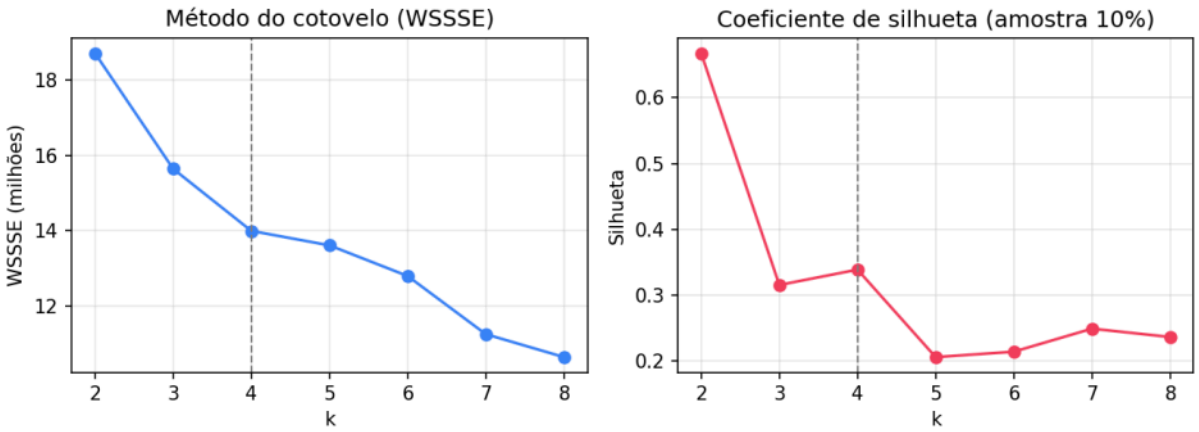


Figure 1 — Evaluation of the number of clusters: elbow (WSSSE) and silhouette for k from 2 to 8.

The silhouette is highest at k=2 (0.67); however, that solution merely distinguishes, in a coarse way, the acoustic/older sound from all the remaining observations, offering little analytical value. The WSSSE decreases sharply up to k=4, levelling off from that point onwards (elbow), and the silhouette recovers slightly at k=4 (0.34) compared with k=3 (0.32). Favouring interpretability over the numerical optimum, K=4 was set, the smallest number of groups capable of producing distinct and nameable profiles. This is a reasoned and discussed decision, not an arbitrary one.

Assessment of the relationship with time. Once the clustering was complete, the groups were cross-tabulated with the decade (contingency table) and the strength of the association was measured using Cramér's V. This indicator was chosen instead of the chi-square p-value, since replication by weights inflates n (around 2 million), making any p-value artificially significant; Cramér's V is an effect size measure that is robust to sample size.

4. Results

4.1 Four sound profiles

K-means with K=4 identifies four clearly interpretable archetypes. Table 1 summarises the means (weighted, on the original scale) of the main features per cluster.

Cluster (profile)	Dance.	Energy	Acoust.	Instrum.	Valence	Tempo
C0 — Pop	0.73	0.68	0.16	0.01	0.73	112
C1 — Upbeat Music	0.54	0.75	0.14	0.01	0.53	136
C2 — Instrumental	0.62	0.62	0.31	0.70	0.65	120
C3 — Acoustic / orchestral	0.52	0.40	0.57	0.01	0.47	114

Table 1 — Mean (weighted) profile of each cluster on the most distinctive features.

- **C0 – Pop (dominant).** High danceability and valence values and intermediate energy; corresponds to the typically commercial sound, geared towards radio airplay. It is the largest group (859 thousand weighted observations).

- **C1 – Upbeat music.** High energy and pace (136 BPM) and lower valence and danceability values; it covers rock, electronic music and more energetic pop.
- **C2 – Instrumental.** Stands out for very high instrumentalness (0.70), unique among the groups; it brings together instrumental tracks. It is the smallest group (57 thousand) and is in decline.
- **C3 – Acoustic / orchestral.** Very high acousticness (0.57) and low energy and loudness values — it covers ballads and the acoustic/orchestral sound characteristic of the early decades.

4.2 Evolution of the profiles across the decades

The cross-tabulation with the decade clearly answers RQ2 (Figure 2). The association is moderate and statistically robust: Cramér's V = 0.28 (chi-square ≈ 484 644; weighted n = 2 053 341). The profiles are not evenly distributed over time; on the contrary, they describe a clear transition.

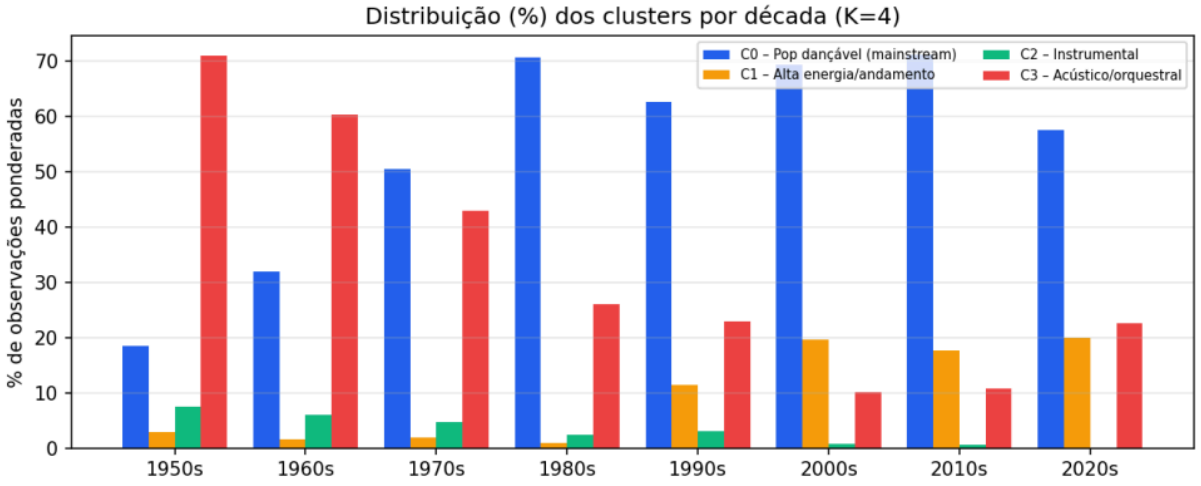


Figure 2 — Distribution (%) of each cluster by decade (each decade sums to 100%).

In the 1950s and 1960s, the C3 acoustic/orchestral profile predominates (71% in the 1950s and 60% in the 1960s). As the decades progress, this profile recedes almost monotonically (to around 10% in the 2000s and 2010s), giving way to the C0 pop profile, which becomes predominant from the 1980s onwards (60–71%). The C1 Upbeat music profile is practically residual until the 1980s, increasing from the 1990s (12%) and reaching around 20% in the most recent decades. The C2 instrumental profile, always a minority, disappears almost completely (0% in the 2020s). There is also an unexpected result: in the 2020s, the C3 acoustic profile recovers (23%), which may reflect the recent trend of acoustic pop and indie, as well as the still small number of observations from that decade.

Summary of results. Hits are not sonically homogeneous: they are organised into four stable and interpretable profiles (RQ1), and their prevalence traces a coherent trajectory through the history of popular music — from the acoustic/orchestral sound of its origins towards an increasingly danceable, produced and energetic sound (RQ2).

5. Discussion

Plausibility. The patterns obtained are musically plausible and independent of any prior labelling: the decline of the acoustic sound and the rise of the danceable and produced sound go hand in hand with electrification, studio production and the dance aesthetic of recent decades. The fact that the structure emerges exclusively from the audio features reinforces its credibility.

Answer to the questions. Both research questions receive an affirmative and well-founded answer: natural profiles exist (RQ1) and their composition evolves systematically over time (RQ2), with a moderate association (V = 0.28).

Stability and ambiguity. Not all results are equally clear-cut. The silhouette at K=4 (0.34) indicates moderately overlapping groups rather than well-separated spheres, which is to be expected in music data, where the boundaries between styles are continuous. The boundary between C0 (danceable) and C1 (Upbeat music) is the least stable, since both correspond to modern sounds and their exact separation depends on initialisation. By contrast, the extreme profiles, C3 acoustic and C2 instrumental, prove robust and easily recognisable.

Limitations. Several decisions condition the results and should be read with caution:

- **Weighting and inference.** Replication by weights inflates n and invalidates the chi-square p-value, which is why Cramér's V was reported. The choice of weighting function (square root) is a modelling decision that conditions the centroids.
- **Dataset bias.** The Hot 100 covers only hits from the United States, so the profiles describe North American commercial success and not music in general. The Spotify features were obtained a posteriori and may contain errors (anomalous cases were detected in Phase 1).
- **Loss of information.** Reducing each song to ten means, followed by standardisation, removes nuances; in addition, the "song-week" unit favours longer-lasting hits.
- **Parameter dependence.** K=4 is a compromise solution; other values of k would produce different readings, although the overall temporal trend would remain.

6. Conclusions

The unsupervised analysis of the audio features of Billboard hits revealed four interpretable sound profiles, pop, high energy, instrumental and acoustic/orchestral, and showed that their prevalence evolves systematically over the decades (Cramér's V = 0.28). The main contribution of the work lies in translating a large dataset into a clear reading: hit music moved from an acoustic/orchestral sound to a danceable and produced sound, with the high-energy strand emerging from the 1990s onwards.

The main limitations lie in the moderate separation between the two modern profiles, the bias of the dataset towards the North American market and the dependence on pre-processing choices (weighting and K). As future extensions, the following are suggested: the application of density-based clustering methods or Gaussian mixture models, which better tolerate group overlap; the assessment of the stability of the

solution through resampling (bootstrap); and the enrichment of the unit of analysis with the length of stay in the chart or with the musical genre.

7. Statement on the use of AI tools

In the development of this work, generative AI tools such as Google Antigravity and Claude Desktop were used as support, namely for:

- the initial conceptual clarification of clustering evaluation metrics (silhouette, WSSSE and Cramér's V);
- suggestions on the structure and organisation of the PySpark code and its comments;

8. References

- Billboard Hot 100 — dataset «Billboard Hot weekly charts» (Data.World), weekly appearances 1958–2021.
- Spotify Audio Features — audio attributes obtained via the Spotify Web API (Audio Features endpoint).
- Apache Spark — MLlib: KMeans, StandardScaler, PCA, ClusteringEvaluator. Official documentation spark.apache.org.
- Rousseeuw, P. J. (1987). Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. J. Comp. Appl. Math.
- Cramér, H. (1946). Mathematical Methods of Statistics. Princeton University Press.
- Libraries: PySpark, pandas, NumPy, SciPy (chi2_contingency), Plotly.

9. Appendices

Appendix A — Evaluation metrics by k

k	WSSSE	Silhouette (10% sample)
2	18 717 837	0.667
3	15 646 821	0.315
4	13 997 227	0.339
5	13 613 391	0.206
6	12 798 639	0.214
7	11 253 307	0.249
8	10 643 856	0.236

K=4 chosen as a balance between the WSSSE elbow and interpretability.

Appendix B — Distribution (%) of clusters by decade

Decade	C0	C1	C2	C3
1950s	18.5	2.9	7.6	70.9
1960s	32.0	1.7	6.0	60.3
1970s	50.5	1.9	4.7	42.9
1980s	70.5	1.0	2.4	26.1
1990s	62.5	11.5	3.1	22.9
2000s	69.2	19.7	0.8	10.2
2010s	71.0	17.7	0.6	10.8
2020s	57.4	20.0	0.0	22.6

Appendix C — Reproducibility notes

- Script: PySpark_MPL09.py (kmeans_hits). Data expected in /BigData/Projecto/data/ (billboard.csv, audio_features.csv).
- Fixed random seed (seed=42) in K-means, in the silhouette sampling and in the PCA projection.
- Data files not included in the submission, as per the assignment brief.